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Symptomatic dataset For Autism Spectrum Disorder in Arab Children

Dina Ayman Abu Taleb
Ain Shams University
Mohmed Mabrouk Morsey
mohamed.mabrouk@cis.asu.edu.eg

Ain Shams University
Manal Omar
Ain Shams University
El-Sayed M. El-Horbaty
Ain Shams University

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Abstract

Autism spectrum disorder (ASD) is a neurological disorder that affects the ability of communicative, linguistic and social skills. Early detection of ASD especially on children is important and affect the quality of the life that the children will live in the future. Recently, there are many techniques used to build models that can predict ASD early depending on the data. This paper builds a dataset for ASDs with focus on Arab children “Egyptian” to be used by researchers in research that depends on data. The methodology used Modified Checked list for Autism in toddlers, revised(M-CHAT-R) as a tool to collect data which is a behavioral test. The result from the collection process, is a dataset file consisting of 200 patients, some of them have ASD, ASD with other syndromes, and others syndromes that have some symptoms similar to ASD.

Introduction

Autism Spectrum Disorder (ASD) results from a neurological disorder that affects the brain's functions [1]. Approximately 10% of ASD cases have a defined etiology, while 90% are idiopathic without a definite causative agent. Many experts in ASD field consider that the interaction between genetic and environment factors play a role in ASD [2]. Screening and surveillance are complementary in the diagnosis process for autistic children in which it starts with screening using questionnaire or other tools to identify if the child needs a further assessment or not. On the other hand, surveillance depends on a continuous process with the child’s parents to obtain a full picture about the developmental health over time [3]. After the diagnosis process is done the doctor or clinical decides if the child has ASD and needs special care or not. The previous process is a time consuming, inefficient, and complex which delay the diagnosis of the child and leads to delays in starting of the care provides to autistic child.

Nowadays, various data-driven approaches are utilized to enhance prediction and diagnosis processes. Advanced analytical techniques help improve the accuracy of pre-diagnostic classifications, enhance efficiency, and increase accessibility to screening methods [13]

For this purpose, dataset plays an important role in developing predictive models for ASD, as they provide necessary information to train these models. The datasets available are wide in their type of data they include clinical information, images, demographic and behavioral assessments.

Despite the variety of datasets, there is a challenge in data quality, data sample size, ethical and cultural bias, data imbalanced, Missing data, limited features, age-related bias, lack of standardization, and subjectivity in diagnoses. All these challenges can make the model that developed using the datasets not generalized and poor in predicting and detecting ASD.

In recent years, the application of machine learning and data-driven approaches in ASD research has gained significant attention. Numerous studies have developed predictive models for ASD using diverse datasets. Comprehensive datasets encompassing the breadth of ASD symptomatology enable

researchers to construct more robust models. When trained on high-quality datasets, these models can enhance screening protocols, identify risk factors, and delineate potential ASD subtypes.

Several key resources for ASD-related datasets include:

- **Autism Data from UCI Machine Learning Repository** provides several datasets related to ASD that are suitable for classification and predictive modeling tasks. These datasets cover various age groups, including adults, children, and adolescents, and contain attributes that facilitate machine learning research on ASD screening. Each dataset includes key features like demographic and behavioral data, allowing researchers to develop models for early autism detection. The repository is widely used for educational and research purposes in the field of machine learning [4].
- **Kaggle** website offers a wide range of datasets related to ASD, making it an excellent resource for machine learning and data science projects. The available datasets include attributes such as demographic information, behavioral screening results, and medical history, allowing for diverse analysis and predictive modeling. Kaggle also provides tools for collaboration and sharing research insights, further enhancing its utility in ASD-related research [5].
- **Autism Brain Imaging Data Exchange (ABIDE)** it is one of the largest open datasets for ASD, providing neuroimaging data (MRI, fMRI) from individuals with and without autism. It aims to foster the development of new algorithms and methods for analyzing brain activity in ASD. These datasets enable researchers to explore neural mechanisms underlying ASD and develop machine learning models for improved diagnostic and therapeutic interventions [6].
- **National Database for Autism Research (NDAR**) is the fourth repository that provides a large-scale data repository funded by the NIH. The dataset for ASD offers valuable information for healthcare providers, and researchers which can be utilized for statistical analysis, predictive modeling, and healthcare planning [7]. These dataset repositories are among the most prominent and applicable in research related to ASD.

Some researchers focus on a specific age group of individuals with ASD and utilized a single dataset. For instance, Sumia et al. [8] and Koushik et al. [9], who employed a dataset of adults from the UCI repository to construct a prediction model. On the other hand, many researchers used more than one dataset. Rownak et al. [10] used 3 datasets from UCI repository for different age group children, adult, and the third one they used by combined the children and adult datasets to build a model that can diagnosis ASD early.

In the same point, Trapti et al. [11] used four datasets for toddler, children, adult, and adolescent from UCI repository and Kaggle to study the challenges of building a prediction model for ASD.

Table 1
Comparative between datasets that are commonly used

Dataset Name	Data type	Sample size	Accessibility	Data format	Repository
Autism Screening Adult	Behavioral, Demographic	704 with 192 missing values	public	ARFF	UCI repository
Autistic Spectrum Disorder Screening Data for Children	Behavioral, Demographic, Screening Data	292 with 90 missing values	Public	ARFF	UCI repository
Autism screening data for toddlers	Behavioral, Demographic, Screening Data	1054 with no missing values	Public	CSV, Excel file	Kaggle
Autistic Spectrum Disorder Screening Data for Adolescent	Behavioral, Demographic, Screening Data	104 with 12 missing values	Public	ARFF	UCI repository
Autism Spectrum Disorder Symptomatic dataset: For Arab Children	Symptomatic dataset	200 with no missing values	Public upon request	CSV, Excel file	Science data bank

Table 1 represents a comparison between datasets that are commonly used by researchers and the dataset we developed. These datasets are not specific for Arab or middle east patients.

Our dataset prioritizes children with ASD in Egypt focused on North Cairo, as it is an Arab country. Employing the Modified Checked list for Autism in toddlers, the revised (M-CHAT-R) [12] test was used to collect data from children's parents. This dataset will be used to develop a prediction model to predict ASD in toddlers as early as possible so that the child can receive suitable care, which helps improving the life level.

Methods

Sample Collection process

This research focuses on collecting data to build a comprehensive dataset for children with ASD in Egypt, as early diagnosis is crucial. Data was gathered from multiple institutions, including the Ain Shams Center for Special Needs Care, the Egyptian Autistic Society, and Resala Charity Organization, along with interviews with numerous parents of children with ASD. This approach ensures data diversity, quality, and an expanded dataset scope.

The data collection for the ASD dataset was conducted using the M-CHAT-R, a structured behavioral screening test consisting of 20 questions assessing child behavior. Participants on the data collection

process were the parents of the children in North Cairo Governorate – Egypt. The process began at the Ain Shams center of the care of people with special needs which accommodates many autistic children aged 3 to 15 years, after that the M-CHAT-R test applied to autistic children on the Egyptian Autism Society, a key organization providing ASD care in Egypt. Following the same approach, data collection process continued on Resala charity organizations, and additional data was gathered through interviews with parents of autistic children reached via social media platforms such as WhatsApp and Facebook.

The diversity of the places lead to diversity on the sample worked on it from different aspects like: social classes, cultural and educational level, and surrounding environment which are important in the diagnosis, and care process for ASDs.

The study was approved by the Research Ethics Committee of the Faculty of Computer and Information Sciences, the research adhered to the ethical guidelines ensuring compliance with data protection, participant confidentiality, and responsible data sharing. The process started by telling the parents about the aim of the data collection process, and giving them some information about M-CHAT-R test. Before initiating the test, some questions were asked to parents about the child's age, and the first diagnosis they got when they have a doubt that the child is abnormal.

Written Declaration of free and informed consent to participate in scientific research was obtained from all parents, ensuring that they understood the study's objectives, data usage, and confidentiality measures. The consent process adhered to ethical guidelines set by Ministry of Health & Population General Secretariat of Mental health, and participants' identities were anonymized to protect privacy.

To resume the process, parents were asked a series of questions about their child's behaviors and asked them to answer the questions in the past when their child was between 12 and 36 months by given the parent a time to remember if their child did this behavior or not.

To sum up the collection process, the dataset contains 200 patients from different places, 111 of them suffering from ASD, and 12 of the ASDs have a second syndrome alongside ASD, the remaining children are normal or have other syndromes like Mental deficiency, Down syndrome, dyslexia, depression, and ADHD.

Data preprocessing

to ensure the reliability and accuracy of the dataset, preprocessing was applied to the questions used to build the dataset to ensure that the questions are in the core of ASD and give more accurate results when they are used to build a prediction model.

The following Table 2 represents the 20 questions before preprocessing:

Table 2
Test Questions that used to build dataset

No	Questions	Answers
Q1	If you point at something across the room, does your child look at it?	Yes, No
Q2	Have you ever wondered if your child might be deaf?	Yes, No
Q3	Does your child play pretend or make-believe?	Yes, No
Q4	Does your child like climbing on things?	Yes, No
Q5	Does your child make unusual finger movements near his or her eyes?	Yes, No
Q6	Does your child point with one finger to ask for something or to get help?	Yes, No
Q7	Does your child point with one finger to show you something interesting?	Yes, No
Q8	Is your child interested in other children?	Yes, No
Q9	Does your child show you things by bringing them to you or holding them up for you to see – not to get help, but just to share?	Yes, No
Q10	Does your child respond when you call his or her name?	Yes, No
Q11	When you smile at your child, does he or she smile back at you?	Yes, No
Q12	Does your child get upset by everyday noises?	Yes, No
Q13	Does your child walk?	Yes, No
Q14	Does your child look you in the eye when you are talking to him or her, playing with him or her, or dressing him or her?	Yes, No
Q15	Does your child try to copy what you do?	Yes, No
Q16	If you turn your head to look at something, does your child look around to see what you are looking at?	Yes, No
Q17	Does your child try to get you to watch him or her?	Yes, No
Q18	Does your child understand when you tell him or her to do something?	Yes, No
Q19	If something new happens, does your child look at your face to see how you feel about it?	Yes, No
Q20	Does your child like movement activities?	Yes, No

The pervious 20 questions were reviewed with a doctor specialist on Autism, after discussion and deep understanding for each question we delete the questions that do not in the core of ASD or talked about symptoms common with other syndromes, to become 13 questions that represents on Table 3:

Table 3
Questions after preprocessing

NO	Questions	Answers
Q1	If you point at something across the room, does your child look at it?	Yes, No
Q2	Does your child play pretend or make-believe?	Yes, No
Q3	Does your child point with one finger to ask for something or to get help?	Yes, No
Q4	Does your child point with one finger to show you something interesting?	Yes, No
Q5	Is your child interested in other children?	Yes, No
Q6	Does your child show you things by bringing them to you or holding them up for you to see – not to get help, but just to share?	Yes, No
Q7	Does your child respond when you call his or her name?	Yes, No
Q8	When you smile at your child, does he or she smile back at you?	Yes, No
Q9	Does your child look you in the eye when you are talking to him or her, playing with him or her, or dressing him or her?	Yes, No
Q10	Does your child try to copy what you do?	Yes, No
Q11	If you turn your head to look at something, does your child look around to see what you are looking at?	Yes, No
Q12	Does your child try to get you to watch him or her?	Yes, No
Q13	If something new happens, does your child look at your face to see how you feel about it?	Yes, No

The 13 questions shown on Table 3 discussed main symptoms of ASD that distinguish it from other syndromes and give more accurate diagnosis, for example, in the above table question 1 related to sharing intention which is important because the autistics do not share their intentions with anyone. Also, questions 3,4,6,11 have the same reason.

These questions used to build the dataset in addition to some general data about the child like: age, gender, first diagnosis, and the class which represent if the child has ASD or not. The questions answer in the data file were 0 and 1, which 0 is No, and 1 is yes. Also, in class column 0 represent that the child has not ASD in the class, and 1 has ASD.

Results

We built a dataset for children who have ASD in Egypt. The male with autism in the dataset file 96 male which represents a 48%, and female with autism 15 which represents a 7.5%, This leads us to that the number of males who have ASD is higher than females. The dataset also represents syndromes which

are ASD, Normal, ASD with others syndromes like ADHD, and mental deficiency, and other syndromes like slow learning, down's syndrome, and depression.

Conclusion

This paper has presented the building process for autism dataset with addressing the challenges faced through the work. The dataset contains 200 patients from Egypt with providing rich source of behavioral dataset for autism related to Arab regions that can be used by researchers in many researches related to ASD. The data collected from different centers to ensure the variety of sample. M-CHAT- R used as a tool in the data collection process, preprocessing was applied to the test questions to choose the questions that are in the core of ASD and give a high accuracy when used in building prediction models.

Declarations

Data availability

The dataset is publicly available at Science Data Bank, https://www.scidb.cn/en/detail?dataSetId=0b84f15557744486a2d59366716d4f8d&version=V1 [14]. All files verified that can be downloaded correctly.

Author contributions:

Dina Ayman Abu Taleb: Prepared for data collection, analysis, and preprocessing; published the dataset; wrote the manuscript. Mohmed Mabrouk Morsey: reviewed the manuscript. Manal Omar: Facilitated the data collection process, data preprocessing, and reviewed the manuscript. El-Sayed M. El-Horbaty: reviewed the manuscript.

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Competing interests

The authors declare no competing interests.

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